ResearchVoice AI | Solution Design

PHASE 2 • Feature list / five component spec cards (A2) / integration map (A3) / safety and monitoring plans

Architecture decision: ONE integrated n8n workflow with five functional ownership areas. The team reports that the instructor approved this exception to the guideline's five-file default. Owners below must be filled by the actual team; these are functional components, not claims of five separately published workflows.

End-to-end operating sequence

Researcher defines/approves study variables → creates fictional case + dated visit → types data or submits demo voice → transcript checked → schema-bound Gemini extraction → independent QA/alerts/LangSmith → researcher corrects/approves → XLSX delivered through Telegram.

Must-have feature list (eight maximum)

Priority	Must-have capability	Observed implementation
M1	Study creation and variable approval	/new; schema review / approval
M2	Synthetic case + dated follow-up visit	/case, /visit, YYYY-MM-DD
M3	Typed and opt-in voice collection	Telegram; /voiceon /voiceoff
M4	Transcription and explicit confirmation	Groq Whisper; confirmation buttons
M5	Schema-limited extraction + evidence	Gemini JSON; issues / missing flags
M6	Researcher correction and explicit export approval	Name = value; Confirm & Export
M7	Approved XLSX and idempotent delivery	XLSX sent via Telegram; completed visit log
M8	Refusal, independent QA, monitoring and audit checkpoint	Safety checks; judge, alerts, /monitor, LangSmith QA trace

Five proposed component owners

Owner to confirm	Owned component	Shared governance role
[Member 1]	C1 — Study & schema	[Team lead: assign]
[Member 2]	C2 — Intake / transcription / safety	[Integration owner: assign]
[Member 3]	C3 — Structured extraction	—
[Member 4]	C4 — Evaluation / monitoring / alerts	[Evaluation owner: assign]
[Member 5]	C5 — Human approval / export	—

Nice-to-have (outside graded demo): durable database, governed PHI workflow, EHR integration, stable hosted tunnel, organizational roles and access controls, multi-center research management. Not implemented or claimed as production-ready.

C1 — Study & variable schema

A2 WORKFLOW SPEC CARD | Functional owner 1 of 5 | Single integrated n8n deployment

A2 field	Implementation and contract
Owner	[Member 1 — confirm]
Purpose	Define a study and researcher-approved field contract.
Trigger	/new, study description, schema confirmation/approval, /studies
Input	study_name; proposed variable speech/text; researcher/user session; study_id
Steps	Create/select study; capture variable description; extract JSON schema via Gemini; validate field types, units and allowed values; show preview; store only after researcher approval.
AI component	Gemini structured schema parsing; not a clinical decision model.
Tools / data	Telegram, n8n Code, Gemini HTTP API, workflow static data.
Output	study_id; study_name; variables[{name,definition,data_type,unit,allowed_values,validation_rules}]; schema_approved_at.
Human approval	Researcher approves variable schema before use.
Failure behavior	On invalid/ambiguous fields ask for clarification; no active schema until approval.
Risk tier	Medium: wrong schema cascades into records.
Logging / tracing	Approval callback; schema parsing failure; sample normal/edge/unsafe tests to document.
Test cases (planned)	N1 new study with Age/HbA1c; N2 3-variable schema; N3 reselect approved study; EDGE undefined field type; UNSAFE identifying field request. RESULTS PENDING.

Implementation note: boundaries are FUNCTIONAL ownership sections within one integrated n8n workflow, as approved by instructor; not five independently deployed automations.

C2 — Voice/text intake & privacy gate

A2 WORKFLOW SPEC CARD | Functional owner 2 of 5 | Single integrated n8n deployment

A2 field	Implementation and contract
Owner	[Member 2 — confirm]
Purpose	Accept fictional case visits, screen input and make a confirmed transcript.
Trigger	/case, /visit, date, case text or opted-in voice; Telegram Trigger
Input	user_id/chat_id (internal routing), study_id, case_code, visit_number, visit_date, file_id or text.
Steps	Validate synthetic case code and explicit date; screen new text; optionally fetch audio; rename .oga to .ogg; call Groq; screen transcript; show transcript confirmation; route verified text.
AI component	Groq Whisper large-v3-turbo for speech recognition; regex-based safety screening.
Tools / data	Telegram, Groq API, n8n Code, Cloudflare tunnel.
Output	case_id, visit_id, visit_date, pending_transcript, transcript_kind, safety disposition.
Human approval	Researcher confirms transcript; opt-in /voiceon acknowledges external audio processing.
Failure behavior	Reject apparent identifiers/prompt injection/clinical requests; retry transcription or request typed input.
Risk tier	High: raw demo audio reaches Groq BEFORE transcript inspection; prohibit real patient audio.
Logging / tracing	Record synthetic rejection category, STT failures, handoff status; no raw audio in LangSmith.
Test cases (planned)	N1 typed synthetic case; N2 clean voice; N3 follow-up visit; EDGE ambiguous transcription; UNSAFE synthetic identifier pattern. RESULTS PENDING.

Implementation note: boundaries are FUNCTIONAL ownership sections within one integrated n8n workflow, as approved by instructor; not five independently deployed automations.

C3 — Schema-constrained extraction

A2 WORKFLOW SPEC CARD | Functional owner 3 of 5 | Single integrated n8n deployment

A2 field	Implementation and contract
Owner	[Member 3 — confirm]
Purpose	Convert verified research observations into approved structured variables with evidence.
Trigger	Confirmed transcript with approved schema; schema_ai / values_ai route
Input	study.variables, pending_transcript, study_id, case_id, visit_id, visit_date.
Steps	Build restricted prompt; Gemini structured extraction; JSON parse and type validation; verify verbatim evidence; flag conflicts, missing and unknown variables; create draft.
AI component	Gemini 3.5 Flash Lite; JSON-only generation; output not used to diagnose/treat.
Tools / data	Gemini API, n8n Code and study schema in workflow state.
Output	draft.values{}, draft.evidence{}, draft.issues[], draft.missing[], draft.ignored[].
Human approval	No autoexport. Draft sent to component C5 for independent researcher review.
Failure behavior	On failed parse keep confirmed transcript for retry; do not infer missing values.
Risk tier	High: transcription/extraction mismatch could corrupt research records.
Logging / tracing	Record model/parse failure, field-level corrections; synthetic gold-label test cases pending.
Test cases (planned)	N1 one numeric variable; N2 multiple typed values; N3 clear missing-value flag; EDGE contradictory units; UNSAFE synthetic prompt injection attempt. RESULTS PENDING.

Implementation note: boundaries are FUNCTIONAL ownership sections within one integrated n8n workflow, as approved by instructor; not five independently deployed automations.

C4 — Independent QA, monitoring & alerts

A2 WORKFLOW SPEC CARD | Functional owner 4 of 5 | Single integrated n8n deployment

A2 field	Implementation and contract
Owner	[Member 4 — confirm]
Purpose	Assess drafts, flag quality/safety problems and create metadata-only LangSmith records.
Trigger	New values draft after Parse Model Classification
Input	draft fields, schema, confirmed transcript for judge context (model input), visit-scoped evaluation.
Steps	Judge rubric 1–5 across five dimensions; validate JSON; classify soft/hard; log bounded monitoring events; emit Telegram alert when required; POST anonymized QA checkpoint to LangSmith.
AI component	Independent Gemini judge (same model family/provider as extractor, separate request and rubric); NOT clinically calibrated.
Tools / data	n8n judge nodes, /eval, /monitor, LangSmith /runs API, Telegram alerts.
Output	scores{}, rationales{}, average, overall_assessment, monitor_alert_level, monitor_reason_code, judge_duration_ms, qa trace.
Human approval	Hard export hold when factual/safety score <=2 or critical flag; researcher can /redo.
Failure behavior	Judge timeout/invalid output: record failure and soft review warning; tracing transport failure is nonblocking.
Risk tier	High: overreliance on uncalibrated judge; monitoring is not clinical clearance.
Logging / tracing	LangSmith checkpoint (not full per-call trace); 200-event rolling monitoring; 2-person calibration pending.
Test cases (planned)	N1 all scores pass; N2 QA trace accepted; N3 soft alert; EDGE judge invalid JSON; UNSAFE critical-score hold (synthetic controlled test). RESULTS PENDING except known documented examples.

Implementation note: boundaries are FUNCTIONAL ownership sections within one integrated n8n workflow, as approved by instructor; not five independently deployed automations.

C5 — Researcher approval & XLSX delivery

A2 WORKFLOW SPEC CARD | Functional owner 5 of 5 | Single integrated n8n deployment

A2 field	Implementation and contract
Owner	[Member 5 — confirm]
Purpose	Keep final record under human control and deliver researcher-confirmed visit XLSX.
Trigger	Final Review display and Confirm & Export callback, /review, /retry
Input	study/case/visit metadata; draft values/evidence/issues/missing; hard-hold status; approval nonce.
Steps	Show values/missing/issues; accept Name = value corrections; require explicit missing acceptance; enforce hard hold; confirm; create rows; convert XLSX; deliver to Telegram; record completion.
AI component	No autonomous clinical AI; receives extracted draft and judge decision.
Tools / data	n8n Code, Convert to File (XLSX), Telegram Send Document.
Output	XLSX rows with variable, value, unit, evidence, status, flags and confirmation metadata; delivery receipt.
Human approval	Researcher checks every field; export only after explicit confirmation.
Failure behavior	Block unresolved issues/unsafe hold; failed delivery not counted completed; /retry with duplicate caution.
Risk tier	High: wrong export can propagate bad research data; synthetic-only.
Logging / tracing	Approval/export metadata recorded locally; current LangSmith QA checkpoint occurs before final export.
Test cases (planned)	N1 approved two-field XLSX; N2 corrected field export; N3 follow-up visit export; EDGE missing-field acceptance; UNSAFE approval bypass attempt. RESULTS PENDING except known exports.

Implementation note: boundaries are FUNCTIONAL ownership sections within one integrated n8n workflow, as approved by instructor; not five independently deployed automations.

Integration Map | A3

Exact named fields taken from the working integrated n8n workflow; functional handoffs, not separate webhooks.

From → to	Handoff / trigger	Actual field contract
C1 → C2	Schema approved; case begins	study_id, study_name, variables[], schema_approved_at; session study_id
C2 → C3	Confirmed case transcript; action values_ai	source.action; source.user_id; source.chat_id; session: study_id, case_id, visit_id, visit_date, pending_transcript
C3 → C4	Value draft after JSON validation	draft.values{}, draft.evidence{}, draft.issues[], draft.missing[], draft.ignored[]; study.variables[]
C4 → C5	Judge/alerts complete; review displayed	evaluations[visit_id]: status, scores, average, overall_assessment, critical_safety_failure; monitor_alert_level; original review action=send_values
C5 → Telegram	Explicit Confirm & Export	record_id=visit_id; confirmed_at; approved XLSX binary; delivery receipt message_id
C4 → LangSmith	Quality checkpoint created after judging	session_name=ResearchVoice-AI-Graduation; QA summary, scores, alert metadata; no raw text, IDs, audio

Integration and change control

- Only one published Telegram Trigger per bot. No manual copying between components in live execution.
- Schema fields are case-sensitive; the transcription mismatch HCBA1C vs HbA1c was correctly flagged for human review.
- Current state is stored in n8n workflow static data, not a durable database; imports and concurrent runs can affect sessions. Do not treat it as production storage.
- Integration Owner [assign] approves input/output changes and updates this A3 table on the day of change.

Safety Plan

Phase 2 requirement: synthetic source, PHI boundary, risk tiers, refusal/escalation and human approval.

Data and PHI boundary

All demonstration observations are fictional or fully de-identified. A synthetic-only warning appears in Telegram. Typed input and Groq-returned transcripts are screened with best-effort rules for apparent emails, phones, ID numbers, identifying labels, instruction attacks, and requests for clinical decisions. Screening is incomplete and cannot guarantee PHI prevention.

Critical limitation: with /voiceon, AUDIO IS SENT TO GROQ before its words can be screened. Telegram already receives all submitted messages and media before workflow rules execute. Therefore real patient recordings and identifiable patient data are out of scope. External Gemini judge requests also receive synthetic transcript context; LangSmith only receives summary QA metadata in the current implementation.

Proposed component risk tiers

Component	Tier	Rationale / approval
C1 Study/schema	Medium	Researcher explicitly approves definitions.
C2 Intake/privacy	High	External audio; text PHI gate is best-effort.
C3 Extraction	High	Potential wrong fields; researcher checks evidence.
C4 QA/monitor	High	Uncalibrated LLM judge; critical alert holds export.
C5 Approval/export	High	Researcher confirmation required before file delivery.

Refusals, escalation and limits

- Clinical requests (diagnosis, prescribing, changing treatment) → refuse, direct to qualified clinician; do not generate advice.
- Identifier or prompt injection detected → block forwarding to Gemini and request synthetic replacement; note voice may already have reached Groq.
- Unclear/missing/contradictory field → flag and require explicit correction or accepted blank; never silently infer.
- Judge hard criterion: safety or factual_accuracy <=2/5, or critical_safety_failure → hold export; alert researcher; /redo for corrected synthetic input.
- Soft criterion: any dimension <=3, judge failure, or unresolved extraction issue → researcher QA notification. Alerts are QA aids, not certified clinical rules.

Pending sign-off: actual named escalation owner, clinical governance authorization for any future real-data pilot, and independent safety verification.

Evaluation & Monitoring Plan

Phase 2 requirement: judge, LangSmith tracing, dashboard, thresholds and notification owner.

Independent LLM-as-judge rubric (1–5)

Dimension	5 means	Low-score interpretation
Factual accuracy	Extracted values and units match explicit synthetic source	2: unsupported or wrong fact; 1: serious contradiction
Groundedness	All fields have exact source evidence	2: mismatched evidence; 1: invented citations
Safety compliance	No observed PHI leakage/clinical advice/override	2: unsafe output; 1: serious disclosure/override
Format adherence	Schema, types, units, missing handling valid	2: major violation; 1: unusable output
Bias / neutrality	No observed demographic assumptions in this sample	2: unjustified assumption; 1: discriminatory stereotype

Judge scores the draft BEFORE later researcher corrections; a correction does not rewrite the earlier score. LLM judge scores are uncalibrated. Two independent teammates who did not build the evaluated component must assess the same three synthetic scenarios and compare scores. Calibration results and the final rubric belong in Phase 4 Test Report.

Tracing and dashboard

Confirmed today: LangSmith receives one privacy-minimized QA checkpoint for a completed draft and displays judge scores, outcome and timing; `/monitor` shows rolling quality metrics. NOT YET COMPLETE against the guideline: tracing every workflow run, model call, and input/output step. Current LangSmith tracing is a checkpoint, not full end-to-end instrumentation.

Dashboard for the checkpoint: the five Business Brief KPIs with measured actuals; judge average and five subscores; evaluation passes/fails; judge processing duration (NOT full-case latency); counts of safety/escalation alerts; export delivery results. Name a dashboard owner [Evaluation Owner — confirm].

Alert thresholds and responsibility

Level	Trigger	Action / recipient
Soft	Any judge score <=3/5, failed judge, unresolved draft issue	Notify researcher in Telegram; manual review. Evaluation Owner monitors trend.
Hard	Safety <=2/5 or factual <=2/5, or critical safety flag	Pause export for that visit, notify researcher; corrected data re-evaluated via /redo.
Transport	LangSmith HTTP failure	Nonblocking: original researcher review proceeds; failure visible in n8n execution.

Monitoring evidence so far: soft alert fired on transcription/schema mismatch (HCBA1C vs HbA1c); manual correction and XLSX delivered; judge average 4.6/5; LangSmith returned “Run created”. No quantified baseline time, 2-person calibration, or full trace coverage claimed.